

Final Project Report - Three-Page Test Summary

Representation Convergence in Independently Trained Neural Networks: from a theoretical hypothesis to a deployed cross-model embedding adapter.

Core claim. The report tests whether independently trained neural networks can learn compatible internal representations - not identical coordinates, but spaces that can be compared, aligned, and sometimes translated by simple maps. The engineering payoff is a shared vector index in which a small on-device image encoder and a larger server encoder interoperate through one lightweight matrix.

How to read the tests. Each test below asks a different question: CKA asks whether two spaces preserve similar pairwise structure; ridge stitching asks whether one space can linearly predict another; retrieval asks whether the map solves the actual search task; hub and neighbourhood tests ask whether the shared structure is portable, local, or global.

Page 1 - Scientific convergence test: GPT-2 vs Pythia-160M

Test	What the test asks	Main reading
A1 - Activation extraction	Runs 10,000 WikiText-103 passages through frozen GPT-2, frozen Pythia-160M, and a random Pythia control; stores mean-pooled hidden states for every layer.	Creates aligned data.
A2 - CKA analysis	Compares every GPT-2 layer to every Pythia layer using linear CKA, a coordinate-free similarity measure based on which inputs each model places near which.	Learning creates structure.
A3 - Linear stitching	Fits one closed-form ridge matrix per matched layer pair, mapping GPT-2 activations into Pythia activations; held-out R2 measures linearly accessible shared variance.	74-80% shared variance.
Random-weight control	Repeats the same CKA and stitching comparisons against a randomly initialized Pythia model with the same architecture and inputs.	Separates learning from architecture.
Sample-starvation fix	The first run used 1,646 samples and underpowered a 768-dimensional ridge fit; rerunning with 10,000 samples raised mean R2 from 0.434 to 0.737.	Bad power hid alignment.
C.6 direction symmetry	Tests GPT-2 -> Pythia and Pythia -> GPT-2; middle-layer asymmetry largely disappears after metric correction, while layer 0 exposes a real positional-information difference.	Maps are mostly symmetric.
C.7 heat-map audit	Replaces the informal "hot diagonal" reading with measured structure: strong trained-vs-random contrast, a strict L3-L11 depth trend, and hubness that softens raw layer matching.	Structure is real but not simple.

Page-1 conclusion. Experiment A supports the strong quantitative version of the hypothesis for two independent language models: a single linear map translates most matched-layer variance, while random weights fail and CKA confirms the effect is caused by learning.

Page 2 - Practical adapter: MobileCLIP-S1 -> SigLIP 2

Experiment B converts the representational-convergence result into an infrastructure component: a 512x768 matrix that lets phone-side MobileCLIP image vectors behave as native SigLIP vectors in a shared retrieval index.

Test	What the test asks	Main reading
B1 - Paired embeddings	Extracts MobileCLIP image vectors, SigLIP image vectors, and SigLIP text vectors for 4,000 COCO image-caption pairs, then checks that native SigLIP image-caption similarity is in the expected band.	Builds the paired adapter set.
B2 - Ridge adapter	Fits a closed-form linear adapter from MobileCLIP image space into SigLIP image space on a 3,000/1,000 split, without modifying either encoder.	One matrix is enough.
B2 - Procrustes	Tests whether a pure rotation is sufficient; Procrustes collapses relative to ridge, showing the spaces differ in anisotropy and scaling, not only orientation.	Rotation alone fails.
B3 - Retrieval test	Uses SigLIP text queries to retrieve images through the adapted phone vectors, comparing against native SigLIP ceiling and raw no-adapter chance baseline.	R@5/R@10 reach 95-98% of ceiling.
B2-bis - MLP control	Trains a higher-capacity nonlinear MLP on the same objective; it adds essentially nothing beyond the linear map.	Relationship is linear.
B4 - Core ML export	Exports the ridge matrix as a 0.79 MB fp16 Core ML package performing MatMul plus L2-normalization.	Deployable artifact.
B5 - Qualitative sandbox	Runs free-text queries over phone-indexed images, including compositional and negative-control queries, and checks fp16 ranking parity.	Meaning survives translation.
B6 - Interactive demo	Exercises the product loop: upload a mobile photo, index through the phone path, auto-caption, and retrieve in the same list as server-indexed photos.	One index, two tiers.
Appendix B - Pair-free translation	Attempts translation with no paired examples using Gromov-Wasserstein plus CSLS refinement; distribution-level alignment improves, but instance-level retrieval remains at chance.	Shape alone is insufficient here.

Page-2 conclusion. The adapter is both scientifically diagnostic and operationally useful: nonlinearity does not improve it, the no-adapter baseline is chance, and the remaining R@1 gap is attributed to information missing from the small mobile encoder rather than to a weak translator.

Page 3 - Follow-up tests: cross-modal structure, scale, hubs, diagnostics

The appendices turn the original pairwise result into a broader map of what kinds of structure are shared: local neighbourhoods, global linear geometry, isotropy, capacity trends, hub coordinates, and deployment decision thresholds.

Test	What the test asks	Main reading
C.8 cross-modal map	Maps DINOv2 image vectors to bge-m3 caption vectors with no joint image-text training; the result is far above chance but weaker than same-modality alignment.	Cross-modal signal exists.
C.10 scale sweep	Repeats the DINOv2 -> bge-m3 test across DINOv2 small/base/large while keeping rows per dimension fixed.	Alignment rises with capacity.
C.11 isotropy correction	Shows GPT-2 caption space looks retrieval-poor mainly because of collapsed geometry; label-free whitening rescues recall without contrastive training.	Frame matters as much as content.
C.12 unsupervised recovery	Uses geometry alone to recover a partial correspondence between spaces; tuned Gromov-Wasserstein reaches substantial matching above chance but remains weaker than supervised mapping.	Geometry is partly self-identifying.
C.13 shared hub	Builds one 512-dimensional hub over five spaces and trains a caption head on one encoder; the head transfers zero-shot to unseen encoders at about 93-96% of native performance.	Shared coordinates are usable.
C.13.2 cross-lineage transfer	Applies the same hub-trained head to SigLIP 2, an independent lineage, and obtains transfer inside the within-family band.	Not just DINOv2 ancestry.
C.13.3 seven-space hub	Extends the hub to seven encoders spanning image/text and four text objectives; composition works well except when writing into collapsed GPT-2 target space.	Direction and target geometry matter.
C.14 encoder ladder	Tests GPT-2, BERT, SBERT, and bge-m3 as text targets; effective rank predicts raw retrieval, and whitening lifts all arms while leaving a training-objective residue.	Isotropy is necessary, not sufficient.
D.2 decision thresholds	Separates verification, per-query ranking, and global membership detection on the shared index; AUC is near-perfect for verification but lower for global membership.	Same index, different decision problems.
D.3 hubness	Measures whether a few gallery items dominate top-k results; the adapter removes most raw cross-space hubness and restores a healthier neighbourhood graph.	Adapter fixes index geometry.
D.5 certification battery	Runs 47 automated checks for power, row alignment, artifact sanity, target geometry, and cache-key integrity across project artifacts.	Measurements are audited.

Final reading. The report does not prove a universal embedding space for all models. It demonstrates a bounded but strong pattern: independently trained encoders often share recoverable relational structure; some of it is global and linear, some only local, and the difference is exposed by the tests above.

Source: Final_Project_Report(4).pdf. This summary compresses the report into test-level sentences and preserves the report's own distinction between measured findings, hypotheses, controls, and deployment claims.